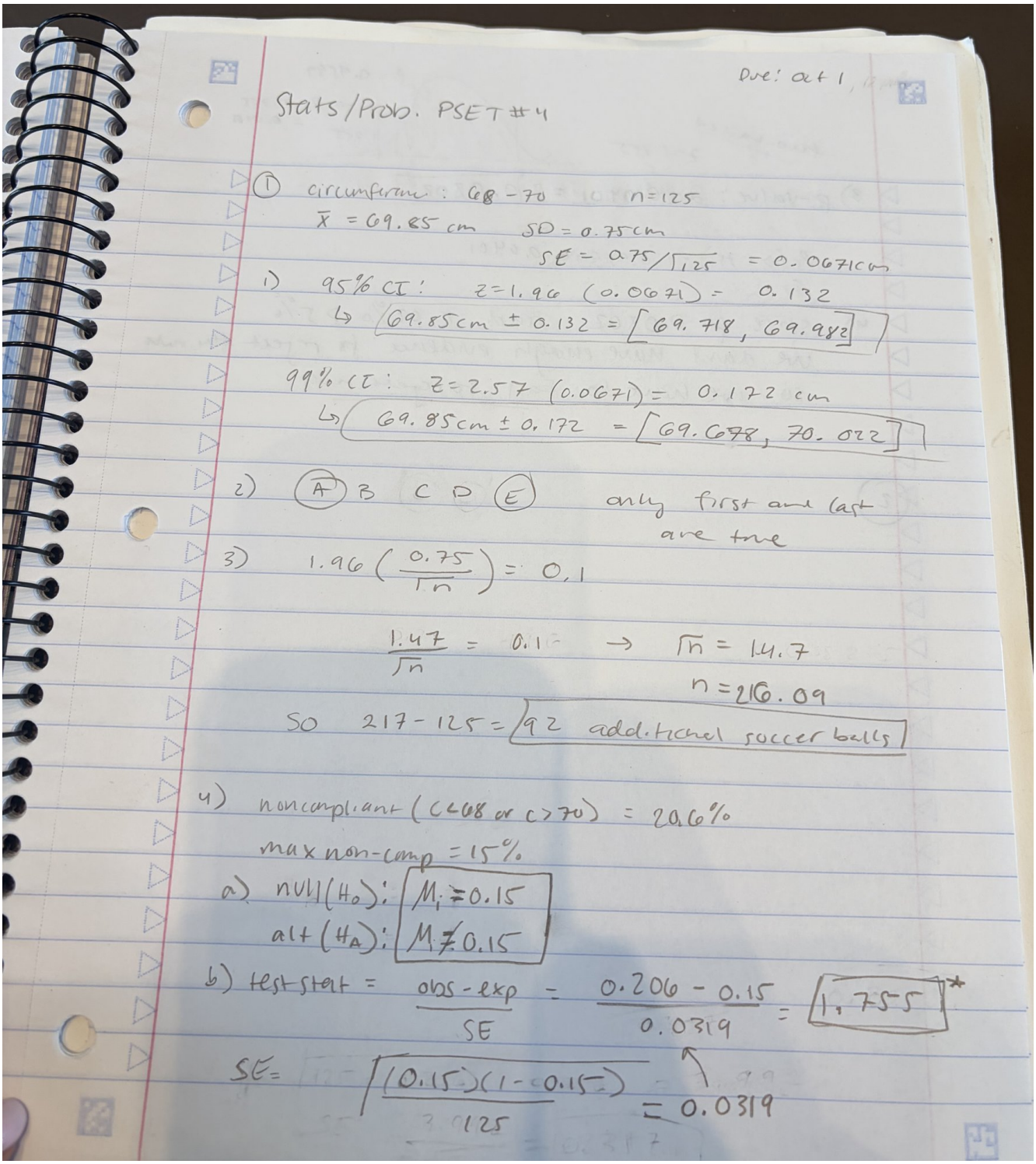
MGT-403 - Problem Set 4

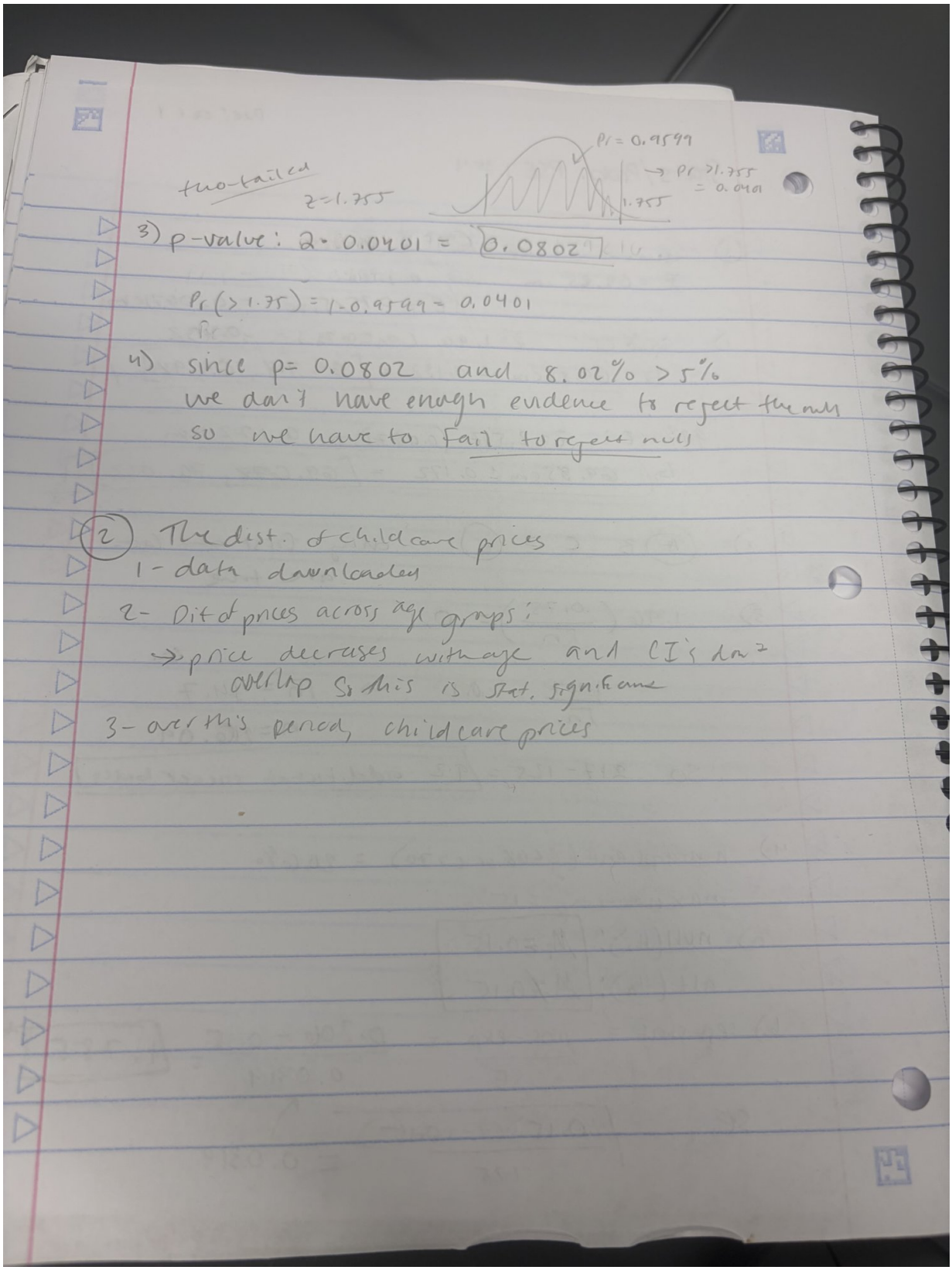
Elise Lehman | September 29, 2026

Part 1 is my handwritten work (photos). Part 2 answers the Data Project questions (Problem Set 4 - The Distribution of Childcare Prices) using the NDCP county workbook (NDCP2022.xlsx), the course-provided county_oews_crosswalk.csv, and BLS CPI-U. All numbers are produced by the Python scripts listed at the end and can be regenerated.

Part 1 - Handwritten work

Handwritten work, page 1 of 2




two-tailed
z=1.755
Pr = 0.9599
→ Pr >1.755
= 0.0401
1.755
3) p-value: 2 · 0.0401 = 0.0802
Pr(>1.75) = 1-0.9599 = 0.0401
4) since p= 0.0802 and 8.02% > 5%
we don't have enough evidence to reject the null
so we have to fail to reject null
(2) The dist. of childcare prices
1- data downloaded
2- Dit of prices across age groups:
→ price decreases with age and CI's don't
overlap so this is stat. significant
3- over this period, childcare prices

Part 2 - Data Project: The Distribution of Childcare Prices

Question 1 - Metro analysis sample

> *Download the NDCP childcare price data. Restrict to the metro analysis sample (counties in an OEWS metropolitan area) so PS4, PS5, and the final project use a consistent county set.*

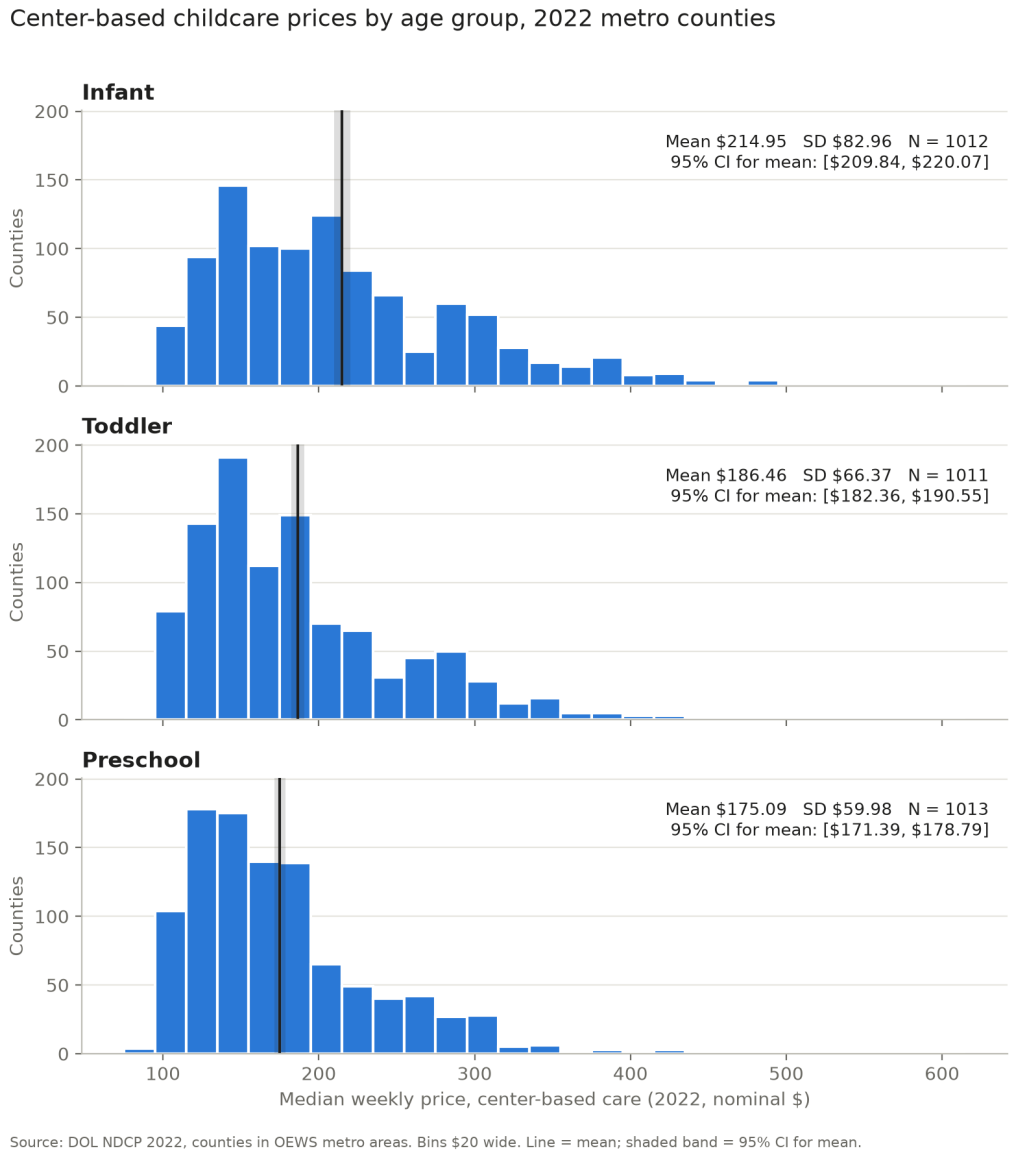
Steps. (1) Read NDCP2022.xlsx with COUNTY_FIPS_CODE as text and zero-pad it to a 5-character FIPS code. (2) Keep study years 2015-2022. (3) Merge on [fips, year] to county_oews_crosswalk.csv, with oews_area kept as a 7-character zero-padded string (the crosswalk is year-specific). (4) Keep counties whose oews_area begins with "00" (metropolitan) and drop non-metro codes. We focus on center-based median weekly prices (MCINFANT, MCTODDLER, MCPRESCHOOL).

Year	Metro counties	With center preschool price
2015	1,233	1,003
2016	1,233	1,025
2017	1,234	1,063
2018	1,233	1,063
2019	1,235	1,063
2020	1,235	1,063
2021	1,235	918
2022	1,235	1,013
Total rows	9,873	8,211

Notes. The metro sample has 9,873 county-year rows and 1,235 counties in 2022, including 69 Puerto Rico municipios (Puerto Rico has center-based prices only in 2021, so it does not affect the 2022 or 2015 analyses). Counties missing a price for an age group are dropped from that age group's statistics only. NDCP price flags (1/2/3) are retained but not used as a filter.

Question 2 - 2022 center-based prices by age group

Using 2022: focusing on center-based care, make histograms of median prices by child age group; report mean / SD / number of observations of prices. Compute 95% CIs for the average value within each age group. Based on your results, discuss differences in prices across age groups.



Source: DOL NDCP 2022, counties in OEWS metro areas. Bins $20 wide. Line = mean; shaded band = 95% CI for mean.

Table 2. County median weekly center-based price, 2022 metro counties (nominal $).

Age group	N	Mean	SD	SE	95% CI for mean	Median
Infant	1,012	$214.95	$82.96	2.61	[$209.84, $220.07]	$198.93
Toddler	1,011	$186.46	$66.37	2.09	[$182.36, $190.55]	$170.00
Preschool	1,013	$175.09	$59.98	1.88	[$171.39, $178.79]	$162.50

95% CI = mean ± $t_{0.975,\ N-1}$ × SD/√N ($t \approx 1.962$).

Table 2b. Within-county price differences, 1,011 counties with all three prices (2022).

Comparison (same county)	Mean difference	95% CI	Share of counties higher
Infant - Toddler	$28.53	[$26.63, $30.43]	92.4%
Toddler - Preschool	$11.31	[$10.14, $12.47]	64.5%
Infant - Preschool	$39.83	[$37.56, $42.11]	96.4%

Discussion - differences in prices across age groups

As children get older, prices fall for childcare, as seen by the means. Also, since the CI's don't overlap, this is statistically significant difference, meaning not by chance. This could be due to lower costs for childcare of older children as compared with younger ones.

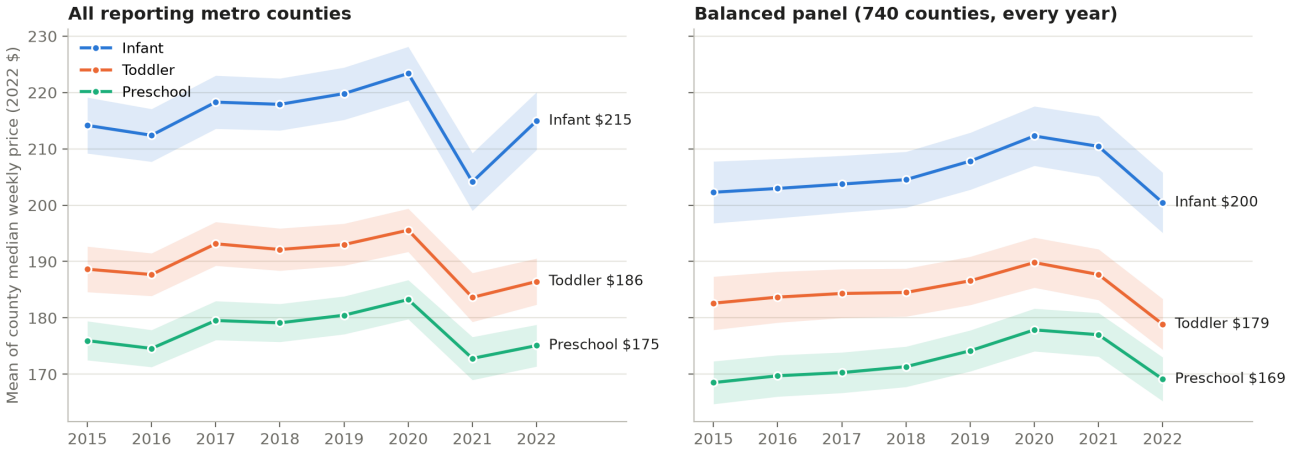
Question 3 - Real prices over time (constant 2022 dollars)

Convert all dollar values to constant 2022 dollars using CPI-U (CUUR0000SA0). Plot trends in median price by year for each age group. Discuss: What has happened to childcare prices over this period?

Real price$_t$ (2022 $) = nominal price$_t$ × CPI$_{2022}$ / CPI$_t$, using CPI-U annual averages (BLS series CUUR0000SA0, U.S. city average, all items, 1982-84 = 100), retrieved from the BLS Public Data API. All six NDCP center-based price columns (median and 75th percentile) were converted.

Year	2015	2016	2017	2018	2019	2020	2021	2022
CPI-U	237.017	240.007	245.120	251.107	255.657	258.811	270.970	292.655
2022 $ factor	1.2347	1.2194	1.1939	1.1655	1.1447	1.1308	1.0800	1.0000

Center-based childcare prices in constant 2022 dollars, 2015-2022



Source: DOL NDCP 2022 (metro counties via OEWS crosswalk); BLS CPI-U CUUR0000SA0 annual averages. Shaded bands = 95% CI for the mean.
2021 in the left panel drops VA, NC, CA, PA counties and adds Puerto Rico (its only year with prices).

The set of counties reporting prices changes by year: 2021 drops counties in VA, NC, CA and PA and is the only year with Puerto Rico prices (about $78/week for preschool), which produces the 2021 dip in the left panel. The right panel is a balanced panel of counties with all three prices in every year, so its trend reflects price changes rather than sample changes.

Table 3a. Mean county median weekly price, 2022 dollars - balanced panel.

Year	Infant	Toddler	Preschool	N (preschool)
2015	$202.29	$182.61	$168.52	740
2016	$202.97	$183.68	$169.72	740
2017	$203.74	$184.33	$170.29	740
2018	$204.53	$184.51	$171.34	740
2019	$207.82	$186.59	$174.16	740
2020	$212.28	$189.82	$177.87	740
2021	$210.44	$187.69	$177.01	740
2022	$200.45	$178.87	$169.13	740
Real 2015-22	-0.9%	-2.0%	+0.4%	
Nominal 2015-22	+22.4%	+20.9%	+23.9%	

Table 3b. Mean county median weekly price, 2022 dollars - all reporting counties.

Year	Infant	Toddler	Preschool	N (preschool)
2015	$214.16	$188.64	$175.96	1,003
2016	$212.40	$187.69	$174.58	1,025
2017	$218.29	$193.15	$179.54	1,063
2018	$217.89	$192.14	$179.12	1,063
2019	$219.80	$193.01	$180.47	1,063
2020	$223.38	$195.56	$183.25	1,063
2021	$204.14	$183.64	$172.81	918
2022	$214.95	$186.46	$175.09	1,013
Real 2015-22	+0.4%	-1.2%	-0.5%	
Nominal 2015-22	+23.9%	+22.0%	+22.9%	

CPI-U rose 23.5% from 2015 to 2022.

Discussion - what has happened to childcare prices?

Nominal prices for childcare over this period rose, however real prices remained fairly flat. Rising a little through 2020 but then decreasing by 2022.

Question 4 - Does the CCDF voucher equal the median center-based preschool price?

> *In 2022 the monthly CCDF voucher for center-based preschool was $706 (2015: $473, nominal). ACF asserts that in the average metro area this voucher equals the price of care at the median-priced center-based preschool. Assuming 4.33 weeks per month, test this at the 5% level.*

(a) Hypotheses. Let μ be the mean, across metro counties, of the county median weekly center-based preschool price (2022 nominal $). The weekly value of the voucher is $706 / 4.33 = $163.05.

$H_0: \mu = 163.05$ versus $H_1: \mu \neq 163.05$ (two-sided, $\alpha = 0.05$)

(b) Test statistic. One-sample t-test:

$t = (\text{x-bar} - \mu_0) / (s / \sqrt{n})$, which follows a t distribution with n - 1 degrees of freedom under H_0.

Element	Calculation	Value
n	counties with a 2022 center preschool price	1,013
x-bar (sample mean)	177,364.65 / 1,013	$175.09
μ_0 (null value)	706 / 4.33	$163.05
x-bar - μ_0	175.09 - 163.05	$12.04
s (sample SD)	$\sqrt{[\Sigma(x - \text{x-bar})^2 / (n - 1)]}$	$59.98
$\sqrt{n}$	$\sqrt{1,013}$	31.8277
SE = s / $\sqrt{n}$	59.9800 / 31.8277	1.8845
t	12.0400 / 1.8845	6.3889
Critical value	t(0.975, df = 1012)	±1.9623

(c) Statistical package and p-value. Computed with Python `scipy.stats.ttest_1samp(x, popmean=706/4.33)` (two-sided). The p-value is the probability, under H_0, of a t statistic at least this far from zero in either direction:

$p = 2 \times P(T_{n-1} \geq |t|) = 2 \times [1 - F_{t,\, n-1}(|t|)]$

	Manual (b)	scipy
t statistic	6.3889	6.3889
p-value	2.54e-10	2.54e-10
95% CI for μ		[$171.39, $178.79]

The manual and package t statistics match (the script asserts equality to machine precision).

Table 4. Test results, 2022 and 2015 (county level, nominal dollars of each year).

	2022	2015
Monthly voucher (nominal)	$706	$473
μ0 = voucher / 4.33 (weekly)	$163.05	$109.24
n	1,013	1,003
Mean weekly price x-bar	$175.09	$142.51
SD s	$59.98	$45.17
SE	1.8845	1.4264
t statistic	6.389	23.324
p-value (two-sided)	2.54e-10	1.84e-96
95% CI for μ	[$171.39, $178.79]	[$139.71, $145.31]
Mean price, monthly (x-bar × 4.33)	$758.13	$617.06
Share of counties priced above voucher	49.0%	75.1%

Robustness: averaging counties within each OEWS metro area (341 areas) and testing the area averages gives t = 4.33 (p = 2.0e-05) in 2022 and t = 15.59 (p = 9.9e-42) in 2015.

(d) Interpretation - do you reject H0, and why? What happens with 2015 data and subsidy levels?

I reject H_0 because the test stat (6.39) in 2022 elicits an extremely small p-value. Because the t stat is about one fourth of the t stat from 2015, meaning that the voucher was much lower than the average price. I also reject the 2015 null hypothesis due to the t-stat.

Note on AI use

I used Claude Code (Anthropic's AI coding assistant) for the data portion of this problem set, as the assignment allows. For Question 1, I gave Claude the assignment PDF and the NDCP2022.xlsx workbook and asked it to read "The Data" section, then restrict the data to 2015-2022 and to the metro analysis sample (counties in an OEWS metropolitan area, per Geography & crosswalks). Claude wrote ps4_step1_metro_sample.py, which zero-pads the FIPS codes, merges the year-specific county_oews_crosswalk.csv, and keeps counties whose oews_area begins with "00".

For Questions 2-4, I gave Claude the assignment PDF and asked for one question at a time. Claude wrote and ran the Python scripts (histograms and 95% CIs, CPI-U conversion and trend plots, and the CCDF voucher t-tests), retrieved the CPI-U annual averages from the BLS Public Data API, and assembled this PDF. It also pointed out data issues (county coverage changing across years, Puerto Rico, the unverified price flags) and suggested the balanced-panel and metro-area robustness checks.

The handwritten work in Part 1 is my own. I checked the outputs, and every result can be reproduced by running the scripts listed below.

Appendix - Code and reproducibility

Run from the PS4 folder, using the project virtual environment in the parent folder (pandas, numpy, scipy, matplotlib, reportlab). The raw inputs (NDCP2022.xlsx, county_oews_crosswalk.csv) sit in the parent folder.

```
python ps4_step1_metro_sample.py ../NDCP2022.xlsx ../county_oews_crosswalk.csv
python ps4_step2_histograms_2022.py
python ps4_step3_real_trends.py
python ps4_step4_ccdf_voucher_test.py
python build_ps4_pdf.py
```

Script	Purpose	Main outputs
ps4_step1_metro_sample.py	Q1: 2015-22, FIPS padding, OEWS metro filter	ndcp_2015_2022_metro.csv
ps4_step2_histograms_2022.py	Q2: histograms, mean/SD/N, 95% CIs	ps4_q2_histograms_2022.png, ps4_q2_summary_2022.csv
ps4_step3_real_trends.py	Q3: CPI-U deflation, trends	ndcp_2015_2022_metro_real.csv, ps4_q3_*.png/.csv
ps4_step4_ccdf_voucher_test.py	Q4: one-sample t-tests	ps4_q4_voucher_test.csv
build_ps4_pdf.py	Assembles this document	PS4_Lehman.pdf